Longitudinal Spin Relaxation of Optically Pumped Rubidium Atoms in Solid Parahydrogen — Supplementary material

Sunil Upadhyay,[1] Andrew N. Kanagin,[1] Chase Hartzell,[1] Tim Christy,[1] W. Patrick Arnott,[1] Takamasa Momose,[2] David Patterson,[3] and Jonathan D. Weinstein[1,*]

[1] *Department of Physics, University of Nevada, Reno NV 89557, USA*
[2] *Department of Chemistry, The University of British Columbia, Vancouver, British Columbia V6T 1Z1, Canada*
[3] *Department of Physics, Harvard University, Cambridge MA 02138, USA*

APPARATUS AND PARAHYDROGEN CRYSTALS

To convert orthohydrogen to parahydrogen, the hydrogen must be held at low temperatures. However, in the absence of a catalyst, the thermalization time is extremely long. Typically, systems to create parahydrogen use two cryostats: one for converting in the presence of a catalyst, and a second for crystal growth, often with a clean room-temperature vessel to store the purified parahydrogen prior to crystal deposition [1, 2].

With the spread of closed-cycle cryostats, techniques have been developed to grow crystals without liquid helium or a storage vessel, using one cryocooler for ortho-para conversion and a second for crystal growth [3]. In our work, we have replaced the two closed-cycle cryocoolers of Tam and Fajardo [3] with a single closed-cycle pulse-tube system [4].

Our ortho-para converters each consist of a $\sim$ 20 cm copper tube which contains roughly 2 cm^3 of iron oxide catalyst [5]. We have two converters. The first is thermally anchored to the first stage of the pulse tube cooler — which has a large cooling power — at a temperature of roughly 50 K. The second is thermally anchored to the second stage of the pulse tube cooler [6]. The second converter is connected through a weak thermal link so that it may be heated to elevated temperatures (to avoid freezing the hydrogen) without significantly raising the temperature of the second stage. After passing through the second converter, the molecules spray onto the sapphire window on which the crystals are grown.

The sapphire window is thermally connected to a copper plate with indium; the copper plate is cooled by the second stage of the cryocooler. The base temperature of the copper plate is 2.7 K. Typical flow rates of hydrogen are 1×10^{18} molecules per second, which results in a crystal growth rate of roughly 3 μm per minute, as monitored by thin-film interferometry [7]. The increase in the temperature of the copper plate during deposition is $\sim$ 0.1 K.

Once a parahydrogen crystal is grown, IR spectroscopy is used to measure its purity, as shown in Fig. 1. The transmission of the crystal is $T = e^{-\mathrm{OD}}$, and is obtained by comparing the spectra of light transmitted through the apparatus before and after growing the crystal. We attribute the negative values of OD measured off-resonance to a combination of lensing by the crystal concentrating the light on the detector and the hydrogen film serving as an antireflection coating for the sapphire substrate.

Fig. 1 shows the spectra of two crystals, one grown with the second ortho-para converter at an elevated temperature of 30 K, and the other grown at its base temperature. The base temperature ranges from 20 K to 24 K during deposition. This could be lowered in future work by increasing the thermal conductivity of the thermal link to the second stage or by better shielding the converter from blackbody radiation. Because we flow hydrogen through the converter at pressures less than 20 Torr, we should be able to operate at temperatures as low as 13 K without freezing the hydrogen in the converter [8].

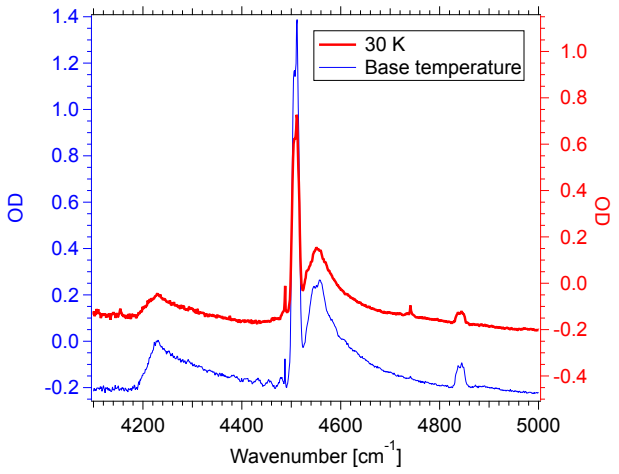


FIG. 1. IR spectra of parahydrogen crystals grown at an elevated ortho-para converter temperature of 30 K, and at our base orth-para converter temperature, plotted against the right and left axes, respectively. The axes are offset to distinguish the curves.

We determine the ortho-para ratio from the relative heights of the spectral peaks at 4740 cm^{-1} and 4490 cm^{-1}. Calibrating the spectrum with the data of Tam and Fajardo [3], we calculate an ortho percentage of $3.2 \pm 0.6\%$ at 30 K and $0.8 \pm 0.3\%$ at our base temperature. These purities are consistent with what we would

* weinstein@physics.unr.edu; http://www.physics.unr.edu/xap/

expect for achieving thermal equilibrium at the measured converter temperatures [9].

The sapphire window is surrounded by a shield thermally anchored to the first stage of our cryocooler to reduce blackbody radiation. Placed outside the 50 K shield is an oven which is used to produce gas-phase rubidium, which passes through a 0.5" diameter hole in the shield. The oven is mechanically connected to — but thermally decoupled from — the 50 K shield [4, 10]. The oven is filled with 99.75% purity rubidium metal. Typically we begin depositing hydrogen with little or no flux of Rb atoms, and then raise the oven temperature until the desired flux of rubidium is obtained.

Typical deposition times are on the order of hours, with typical crystal thicknesses on the order of 1 mm; the thickest crystal grown was roughly 2 mm thick. The IR spectra of rubidium doped parahydrogen is, to within our signal to noise, identical to the undoped spectrum shown in Fig. 1.

We note that when probed in the visible, our crystals exhibit broadband light scattering, even in the absence of Rb atoms. For example, we attribute the nonzero background OD in Fig. 1 of our main paper (roughly 0.5 from 430–540 nm, and considerably less at 970 nm) to be due to scattering by the crystal. Over a timescale of multiple days, the crystal becomes "frosty" as it sits in our apparatus at a temperature of 2.7 K, and this background scattering slowly increases. This background scattering can be reduced by annealing the crystal at 4.2 K for a timescale of hours, typically to levels of OD $\lesssim$ 0.1. The annealing typically has negligible effect on either the visible spectrum or the optical pumping behavior.

[1] T. Momose and T. Shida, Bulletin of the Chemical Society of Japan **71**, 1 (1998).

[2] B. A. Tom, S. Bhasker, Y. Miyamoto, T. Momose, and B. J. McCall, Review of Scientific Instruments **80**, 016108 (2009).

[3] S. Tam and M. E. Fajardo, Review of Scientific Instruments **70**, 1926 (1999).

[4] A. N. Kanagin, *Creation and Analysis of Para-Hydrogen Crystals*, B.S. Thesis, University of Nevada, Reno (2015).

[5] S. Aldrich, Iron (III) Oxide; hydrated, catalyst grade, 30-50 mesh.

[6] Cryomech, PT405.

[7] M. Perera, B. A. Tom, Y. Miyamoto, M. W. Porambo, L. E. Moore, W. R. Evans, T. Momose, and B. J. McCall, Optics letters **36**, 840 (2011).

[8] J. Ekin, *Experimental Techniques for Low-Temperature Measurements: Cryostat Design, Material Properties and Superconductor Critical-Current Testing: Cryostat Design, Material Properties and Superconductor Critical-Current Testing* (OUP Oxford, 2006).

[9] I. F. Silvera, Rev. Mod. Phys. **52**, 393 (1980).

[10] C. Hartzell, *Matrix Isolation of Rubidium in a Solid Para-Hydrogen Substrate*, B.S. Thesis, University of Nevada, Reno (2014).